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Week 01: Wednesday, AST 5011: Astrophysical Systems

Course Introduction and Hydrostatic Equilibrium, Energy Principles, and the Virial Theorem

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

Logistics

- WAYS TO REACH ME
 - Office Hours: Friday 1-2 pm (see Canvas for latest)
 - GitHub: https://github.com/mcoughlin/ast5011_2026_Spring (everything course related goes here)
 - Email: Good for official-ish messages like "I'm missing class/a test."
 - Slack (see Canvas for course invite)
- WAYS TO GET HELP
 - Resources in the syllabus!
 - come to office hours!
 - stackoverflow (can be toxic)

RULES OF ENGAGEMENT

- Being able to explain a concept is the ultimate test of understanding, and it's entirely possible to believe we have an understanding right up to the moment we try to explain it to someone else.
- Therefore, asking questions during class is providing a service.
- No feigning surprise: To enable everyone to be comfortable saying "I don't understand", please resist the urge to act surprised when someone admits to not knowing something.
- No well-actually's: A well-actually happens when someone says something that's almost - but not entirely - correct, and you say, "well, actually..." and then give a minor correction. This is especially annoying when the correction has no bearing on the actual conversation.
- No subtle-isms: It goes without saying that overt expressions of racism, sexism, homophobia, transphobia or other biases are unacceptable. We also need to avoid subtle expressions of bias

that can make others uncomfortable and that are of no benefit to the learning environment. For example, the expression "It's so easy my grandmother could do it" is a subtle-ism.

What are we here for?

BY THE END OF THE CLASS, YOU SHOULD HAVE A BASELINE UNDERSTANDING OF STELLAR STRUCTURE AND EVOLUTION and GALAXIES.

- This class will not replace dedicated classes on these topics, but the goal is to provide an introduction.
- Because it is me, it will also have a software bent... sorry.

Introductions

1. Name
2. Current research interest
3. What you would like to learn in this course
4. Hobbie(s) outside of astronomy

Syllabus overview and discussion

Navigate to the course syllabus [here](#).

Course overview and expectations

Computing

We will be using computing in the following ways.

Modules for Stellar Astrophysics (MESA): [more info here](#).

Jupyter Notebooks: [link to docs](#)

Collaborative working environment

- group work often
- collaboration on assignments, projects, and reviews
- helping each other with various aspects of computing!

My asks

- Be kind
- Be patient
- Be involved

You will succeed in this course! There is room in the schedule to add (or remove) different topics as we progress.

Why github?

- Reproducibility
- Version control (git)
- Collaborative platform

More help with git/os commands/python in the help/directory

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We're not focusing on git/python/os usage - pick it up as we go.

- What the class is NOT
 - A Statistics Class
 - A Math Methods Class
 - A Computer Science Class
 - A Programming Class

Why python?

- intuitive and readable
- open source
- support C integration for performance
- packages designed for science:
 - scipy
 - numpy
 - sklearn
 - matplotlib

Stellar Structure and Evolution

Why Study Stars?

For as long as humans have existed, we have looked up at the stars and told ourselves stories about them. We still do that — but today, those stories are enriched by a remarkable connection between stars and fundamental physics, and by the powerful inferences we can make about the universe through stellar observations.

In the first half of this course, we will tell the story of stars with a focus on:

- The physics governing their birth, life, death, and afterlife
- Developing physical intuition, not just formalism
- Understanding how we know what we know

At any point, it is appropriate — and encouraged — to ask:

How do we know this?

Sometimes the answer comes from basic physics. Sometimes it comes from observations interpreted through theory. And sometimes the honest answer is:

This is our best current guess.

Asking the question is a key to wisdom.

Developing Astrophysical Reasoning Skills

A major transition occurs between solving classroom problems and doing research. In research, you must decide:

- What level of accuracy is actually needed?
- Should this be solved analytically or numerically?
- Is an order-of-magnitude estimate sufficient?

Above all, you must always ask:

Does this make sense?

This question protects you from believing that a programming bug has violated energy conservation.

Common-Sense Checks

Whenever you obtain a result, ask:

- Does it have the right units?
- Does it behave correctly in simple limits?
- Does it respect the relevant symmetries?

- Does it depend on the *right* physical quantities — and no more?

You should ask these questions before and after a calculation.

Creativity in Astrophysics

Creativity often has two stages:

1. Generate possibilities freely
2. Cull them aggressively using physics and observations

Separating these steps helps avoid prematurely discarding good ideas while still enforcing rigor.

Example: What Powers the Sun?

Today we know the answer is nuclear fusion. But imagine we didn't.

Exercise (conceptual):

- What possible power sources could you propose?
- How would you test each one?
- Which observational constraints would rule them out?

This kind of reasoning lies at the heart of astrophysical discovery.

What Do We Know About Stars?

We know an extraordinary amount about stars:

- Masses, radii, luminosities
- Spectra and compositions
- Evolutionary pathways and final fates

But each piece of knowledge comes with a chain of reasoning, often involving decades of refinement and occasional mistakes. Science is a process, not a final answer.

In this course, we will place ourselves in the position of physicists trying to *figure stars out from scratch*.

This means:

- Choosing problems that are simple enough to solve
- But realistic enough to teach us something useful

Deciding What to Ignore: Precision Matters

What you can neglect depends entirely on the precision you need.

A Non-Astronomical Example: A Water Tank

Suppose a water tank has dimensions:

- 107 cm × 109 cm × 103 cm

If you need the mass to the nearest gram, you must know:

- Density variations
- Temperature
- Pressure
- Composition
- Gravity

This is a *very hard problem*.

But if the question is:

Can I lift this tank over my head?

Then an order-of-magnitude estimate is enough:

- Volume $\sim 10^6 \text{ cm}^3$
- Mass $\sim 10^3 \text{ kg}$

Answer: No.

No complexity required.

A Stellar Example: Why Hasn't the Sun Collapsed?

If nothing supported the Sun against gravity, it would collapse on the free-fall timescale:

$$t_{\text{ff}} \sim \frac{1}{\sqrt{G\rho}}$$

For the Sun (average density $\rho \sim 1 \text{ g cm}^{-3}$):

- $t_{\text{ff}} \sim 1 \text{ hour}$

Since the Sun has remained stable for billions of years, something must oppose gravity.

Our first approximation:

- Spherical
- Non-rotating
- No magnetic fields
- Supported by gas pressure gradients

This works remarkably well — but we must justify what we ignore.

Can We Ignore Rotation?

- Solar rotation period: ~ 1 month
- Keplerian period at surface:

$$P_K = 2\pi\sqrt{\frac{R^3}{GM}} \sim 3 \text{ hours}$$

Thus:

- $\Omega_{\odot}/\Omega_K \sim 1/250$
- Centrifugal effects scale as Ω^2

So rotation contributes at the $\sim 10^{-4}$ level.

Unless we need that precision — or rotation introduces qualitatively new physics — we can ignore it.

Can We Ignore Magnetic Fields?

Strongest solar fields (sunspots):

$$B \sim 10^3 \text{ G}$$

If this filled the Sun, the magnetic energy would be:

$$E_B \sim \frac{B^2}{8\pi} \frac{4\pi R^3}{3} \sim 6 \times 10^{37} \text{ erg}$$

Compare with gravitational potential energy:

$$E_{\text{grav}} \sim \frac{GM_{\odot}^2}{R_{\odot}} \sim 4 \times 10^{48} \text{ erg}$$

That's 11 orders of magnitude larger.

Conclusion: magnetic fields are dynamically negligible for global structure — though crucial for flares and X-ray emission.

Types of Equilibrium in Stars

Hydrostatic (Dynamic) Equilibrium

Consider the following assumptions of an object:

- spherically symmetric
- nonrotating
- nonmagnetic
- single star on which there are
- no net forces acting and, hence, no accelerations

Then, we ask the question, how can we describe this "star" in equations?

First, some convections:

- **radius:** r is the radial distance measured from the stellar center (cm)
- **density:** $\rho(r)$ is the mass density at r (g cm^{-3})
- **temperature:** $T(r)$ is the temperature at r (K)
- **pressure:** $P(r)$ is the pressure at r ($\text{dyne cm}^{-2} = \text{erg cm}^{-3}$)
- **mass:** $m(r)$ is the mass contained within a sphere of radius r (g)
- **luminosity:** $L(r)$, the rate of energy flow through a sphere at r (erg s^{-1})
- **local gravity:** $g(r)$ local acceleration due to gravity (cm s^{-2})

Force balance requires:

$$\nabla P = -\rho \mathbf{g}$$

In spherical symmetry:

$$\frac{dP}{dr} = -\rho \frac{GM_r}{r^2}$$

This is one of the four fundamental equations of stellar structure.

Lagrangian Form

equation of hydrostatic (or mechanical) equilibrium (HSE)

$$\frac{dP}{dr} = -\frac{Gm}{r^2}\rho \equiv -\rho g$$

Since $g, \rho \geq 0$, it follows that $\frac{dP}{dr} \leq 0$, and the pressure must decrease from the center outwards, everywhere.

If this equation is violated, local acceleration must occur, the "star" is said to be *out of HSE*.

Brief derivation

We can start by defining the total gravitational potential energy.

the total gravitational potential energy, Ω , of a self-gravitating body is defined as the negative of the total amount of energy required to disperse all mass elements of the body to infinity.

$$\Omega = -q \frac{GM}{R} (\text{ergs})$$

where here, we have performed the integral over r and M and R represent the total mass and radius of the star, respectively. For a uniform density sphere $q = 3/5$.

Next, consider the total internal energy

total internal energy arising from microscopic processes

$$U = \int_m E dm$$

After some math and application of the first and second laws of thermodynamics, we arrive at a similar conclusion regarding the assumptions we first made when we set out.

The equation of HSE written in Lagrangian form (the independent variable is dm),

$$\frac{dP}{dm} = -\frac{Gm}{4\pi r^4}.$$

In fact, we can use Eqn. 1.1 to recover the first definition for HSE.

Thermal Equilibrium

Although the Sun radiates energy, it is locally very close to thermal equilibrium.

Reason:

- Temperature gradient is shallow
- Photons travel only $\sim 1\text{cm}$ between interactions
- Temperature change sampled is $\sim 10^{-4}\text{ K}$

Compared to $T \sim 6000\text{--}10^7\text{K}$, this is negligible.

Thermal Balance

Energy generation must balance energy flow:

$$\frac{dL}{dr} = 4\pi r^2 \rho \epsilon$$

or in Lagrangian form:

$$\frac{dL}{dM_r} = \epsilon$$

This is the energy conservation equation for stellar interiors.

Big Picture

For much of astrophysics, deep insight comes from a simple question:

What is opposing gravity in this system?

Learning to decide what to include — and what to neglect — is one of the most important skills you will develop in this course.

Observables of Stars and Conventions

What We Observe

Directly observable quantities include:

- Flux as a function of wavelength
- Polarization (in some cases)
- Distance (via parallax, e.g., Gaia)

From these we infer luminosity, with corrections for extinction and geometry.

Magnitudes

The magnitude system is:

- Logarithmic
- Inverted (brighter = smaller number)

Five magnitudes correspond to a factor of 100 in flux:

$$\Delta m = -2.5 \log_{10} \left(\frac{F_1}{F_2} \right)$$

- Sirius:

$$m \approx -1.5$$

- Naked-eye limit:

$$m \approx 6$$

Absolute Magnitude

Absolute magnitude is defined as the apparent magnitude a star would have at 10 pc.

The Sun has:

$$M_V = 4.8$$

Photometric Systems

Flux is measured in bands, not infinitesimal wavelengths.

Examples:

- Johnson–Cousins system
- SDSS system

Caution: Magnitudes depend on the chosen filter system.

Spectral Types and Luminosity Classes

Spectral sequence:

OBAFGKM

- Earlier types: hotter, more massive
- Later types: cooler, less massive

The Sun:

- Spectral type: G2
 - Luminosity class: V (main sequence)
-

Hertzprung–Russell Diagram

- Observer's H–R diagram: absolute magnitude vs color
- Theorist's H–R diagram: luminosity vs temperature (increasing leftward)

Key features:

- Main sequence
- Red giants
- White dwarfs

The H–R diagram encodes stellar evolution, populations, and lifetimes.

In-class question: How would interstellar dust move stars on an H–R diagram?

In-Class Exercise

Learning Objectives

- familiarize with symbolic python tools
- use sympy to compute properties of stellar structure
- make estimates of stellar density and comparisons to the Sun

```
{note}
Install sympy via conda using
```
conda install sympy
```
```

Documentation is available [here](#)

IPython console for SymPy 1.14.0 (Python 3.13.2-64-bit) (ground types: gmpy)

These commands were executed:

```
>>> from sympy import *
>>> x, y, z, t = symbols('x y z t')
>>> k, m, n = symbols('k m n', integer=True)
>>> f, g, h = symbols('f g h', cls=Function)
>>> init_printing()
```

Documentation can be found at <https://docs.sympy.org/1.14.0/>

1. Stellar density profile

```
<>:6: SyntaxWarning: invalid escape sequence '\s'
<>:6: SyntaxWarning: invalid escape sequence '\s'
/var/folders/6/_/ttd4h161x6qj1yw7d3rsyr0000gn/T/ipykernel_97861/1554243202.py:6: Syn
taxWarning: invalid escape sequence '\s'
Mstar, Rstar = symbols('M_\star R_\star')
```

Imagine a star has a density profile of the form:

$$\rho(r) = \rho_c \left[1 - \left(\frac{r}{R_\star} \right)^3 \right]$$

where ρ_c is the central density.

$$\rho_c \left(1 - \frac{r^3}{R_\star^3} \right)$$

Find an expression for the central density in terms of $R_\star$ and $M_\star$.

Recall the formulation for the mass equation

$$M(r) = \int_0^r 4\pi r^2 \rho dr$$

1. First, integrate the mass equation to **find** $M(r)$ using [sympys integrate function](#) evaluated from 0 to $R_\star$ (defined above).

$$\frac{2\pi R_\star^3 \rho_c}{3}$$

2. Next, we want to **solve for ρ_c by setting $M_\star$ to the M we computed in 1** using sympys [Solve function](#).

Hint: in sympy you can set define an equality using [Eq relation function](#). For example to set the LHS of an equation to the RHS, you would write `Eq(LHS, RHS)`.

$$\frac{15M_\star}{8\pi R_\star^3}$$

3. Next, confirm that the average density of this star is $\bar{\rho} = 0.4\rho_c$, assuming $\bar{\rho} = M/V$.

Hint: for fractions in sympy you would use the [Rational function](#).

$$\frac{\rho_c}{2}$$

4. Next, use substitution again to plug in the solar radius and mass in cgs and evaluate ρ_c using [evalf](#) to make an estimate for the central density with the equation we found in 2.

3.54041095742568

6. Confirm that $\bar{\rho} \approx 1.4 \text{ (g cm}^{-3}\text{)}$ for this density profile using the numerical value found in 4. and Subs. Similar to the [Sun](#).

1.41616438297027